SYNOPSIS

Objective of the Analysis

The main objective of this work is to evaluate and analyze the indicators and risk profiles of two companies, Value Valves Co., Ltd. (4580.TWO) and Take-Two Interactive Software, Inc. (TTWO) against the market index, S&P 500. The analysis focused on determining the risk and return for each asset and evaluating the diversification effect of including both assets in a portfolio.

Analytical Methods Used

1. **Descriptive Statistics:** Calculation of weekly returns and volatility ranges.
2. **Risk-Adjusted Performance**: The Sharpe ratio was used to measure excess return per unit standard deviation, with Treasury bills used as the risk-free rate.
3. **Inferential Statistics(Hypothesis Testing):** An F-Test for Two Sample Variances was performed to evaluate the presence of a statistically significant difference in the risk (volatility) of the two equities.
4. **Regression analysis:** A regression model was constructed to calculate the beta coefficient (CAMP) for Take-Two Interactive, which allowed us to estimate its systematic risk relative to the S&P 500.
5. **Correlation & Covariance Analysis:** Quantifying the linear relationship between two assets

Key Findings and Results

- **Risk Profiles:** Value Valves was less volatile than Take-Two Interactive, however the F-test was critical in determining the statistical significance of this difference.
- **Market Sensitivity:** In the regression analysis, TTWO had a Beta of 0.116. The R-squared of 0.012 indicates that the S&P 500 accounted for a tiny portion of the stock's weekly price movement, showing that the performance of the stock was mostly due to factors specific to the company during that time period.
- **Diversification:** The correlation coefficient was -0.032, indicates almost no linear relationship between the two. Thus, the assets are great for a diversification strategy to lower unsystematic risk.
- **Performance:** Both assetsshowed negative Sharpe ratios for the analysed period. Value Valves is -0.075 and -0.018 for Take-Two. It indicating that neither asset outperformed the risk free rate on risk adjusted basis.

Reasoning

The analysis is based in MPT, specifically MPT's tenet that investors should be reimbursed for systematic risk and not total risk. Relating to the Analysis, Sharpe Ratio has been described within the context of Risk-Return Trade-off, which is, risk volatility, and return is a separate consideration. Also, using regression to obtain Beta is probably the most simplistic use of the CAPM, which is a common feature in quantitative business texts to teach and prescribe the determination of a rate of return that investors require.